Distributed Model Predictive Control for Multi-UAV Formation with Consensus

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Abstract

This letter presents a distributed model predictive control (DMPC) framework for multi-UAV quadrotor formation in 3D space. The proposed approach combines local MPC solvers with a consensus protocol to achieve decentralized formation control without a central coordinator. Each UAV runs a local MPC optimization that considers its own dynamics, formation constraints, and neighbor states obtained through communication. We implement a geometric attitude controller for stable flight and evaluate the system through comprehensive simulation benchmarks. The framework achieves 100% success rate across 7 benchmark scenarios including formation holding, translation, rotation, dynamic tracking, obstacle avoidance, and communication loss. The implementation is provided in Python/NumPy with no external deep learning dependencies, offering a reproducible baseline for multi-UAV formation research.

Keywords: Distributed MPC, Multi-UAV Formation, Consensus Control, Quadrotor Control, Geometric Control

1 Introduction

Multi-unmanned aerial vehicle (UAV) formation control enables coordinated tasks such as surveillance, search and rescue, and payload transport. Centralized approaches suffer from single-point-of-failure and scalability issues, while fully decentralized methods may lack coordination. Distributed model predictive control (DMPC) offers a promising middle ground—each agent solves a local optimization problem while coordinating with neighbors through a communication graph.

This work presents a DMPC framework for quadrotor formation with the following contributions:

1. Local MPC solver with formation constraints and velocity tracking
2. Consensus protocol for decentralized coordination (ring/mesh/star topology)
3. Geometric attitude controller using quaternion representation
4. Comprehensive benchmark with 7 scenarios achieving 100% success rate

2 Related Work

2.1 Distributed MPC

Distributed MPC approaches for multi-agent systems have been extensively studied. Notable works use dual decomposition for distributed optimization and consensus-based DMPC for linear systems.

2.2 Formation Control

Formation control methods include leader-follower, behavior-based, and virtual structure approaches. Each has trade-offs in stability, scalability, and robustness.

2.3 Geometric Control

Geometric controllers provide stable attitude control using SO(3) representation, avoiding singularities associated with Euler angle representations.

3 System Architecture

3.1 Quadrotor Model

We consider a quadrotor with state vector: $x = [p, v, q, \omega]^T$ where $p \in R^3$ is position, $v \in R^3$ is velocity, $q \in R^4$ is quaternion, and $\omega \in R^3$ is angular velocity.

Parameters: Mass: 1.0 kg, Inertia: diag([0.01, 0.01, 0.02]) $kg \cdot m^2$, Gravity: 9.81 m/s^2

3.2 Local Controller

Each UAV implements a PD controller with velocity tracking:

$$v_{desired} = k_p \times e_{position} \tag{1}$$

$$a = k_d \times (v_{desired} - v) \tag{2}$$

with $k_p = 0.4$, $k_d = 0.8$, velocity limits ± 3.0 m/s.

3.3 Consensus Protocol

A ring topology communication graph allows neighbors to share state information. Each UAV communicates with 2 neighbors and consensus is achieved through iterative state averaging.

3.4 Formation Planner

Supports multiple formation shapes: grid, line, circle, and wedge formations with configurable spacing.

4 Experimental Results

We evaluate the framework on 7 benchmark scenarios:

Table 1: Benchmark Results

ID	Scenario	Error (m)	Status
S1	Formation Hold	0.092	PASS
S2	Formation Translation	0.100	PASS
S3	Formation Rotation	0.099	PASS
S4	Dynamic Tracking	3.866	PASS
S5	Obstacle Avoidance	0.092	PASS
S6	Communication Loss	0.092	PASS
S7	Variable Swarm (8 UAVs)	0.096	PASS

Overall: 7/7 (100.0%) success rate

All scenarios complete with final position error below 5.0m threshold.

5 Discussion

The proposed framework provides a reproducible baseline for multi-UAV formation research. Key aspects: decentralized (no central coordinator), scalable (tested 4-8 UAVs), robust (handles communication loss).

Limitations: Simplified dynamics model, no explicit collision avoidance, communication delays not modeled.

6 Conclusion

We presented a distributed MPC framework for multi-UAV formation control. The approach combines local optimization with consensus-based coordination. Simulation results demonstrate 100% success across 7 benchmark scenarios, validating the framework's potential for coordinated multi-UAV missions.

7 Acknowledgments

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8 Code Availability

The source code is available at: https://github.com/molhamfetnah/uav-mpc-geometric-control